

SAMANTHA GWYNETH B. ARSUA

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SUMMARY

Software Developer with hands-on experience building production-grade backend systems. Developed FinLock, a distributed fintech wallet system with atomic transfers, Redis locking, Kafka event streaming, AES-256 encryption, and OWASP-verified security, deployed live with CI/CD via GitHub Actions.

EXPERIENCE

UI/UX Designer Intern July 2025 – August 2025

Aquila Softwares Corporation, Davao City

- Designed interface components for multiple platform features, ensuring consistency with design systems and usability standards.
- Translated project requirements into structured user flows and interface specifications for developer handoff.

PROJECTS

FinLock: Distributed Wallet System | github.com/samanthagwynetha/Finlock · finlock-api.onrender.com

Java 21 · Spring Boot 3.5 · PostgreSQL · Redis · Apache Kafka · Docker · GitHub Actions · Prometheus · Grafana

- Built a distributed fintech backend (GCash/Maya-style) with JWT auth, multi-currency wallets, and atomic fund transfers between users.
- Implemented Redis distributed locking to prevent race conditions, proved correctness via concurrent curl test (no double-spend, no negative balance).
- Applied idempotency keys, AES-256-GCM encryption at rest, Kafka async event streaming with DLQ retry, and rate limiting via Bucket4j.
- OWASP ZAP scan: 66 checks passed, 0 failures. Full stack containerized with Docker Compose; deployed live on Render with GitHub Actions CI/CD.

DentaCare: Clinic Management System | github.com/samanthagwynetha/dental-clinic-system · dental-clinic-system-three.vercel.app

Next.js 16 · TypeScript · React 19 · Node.js · Express.js 5 · MongoDB Atlas · JWT · Nodemailer · Winston · Jest · Tailwind CSS

- Built a full-stack dental clinic management platform with JWT authentication, Role-Based Access Control (Admin/Staff/Dentist), and a real-time dashboard tracking patients, appointments, and revenue analytics.
- Hardened API security with tiered rate limiting, Helmet.js HTTP headers, and bcryptjs password hashing against brute-force and injection attacks.
- Integrated Nodemailer appointment reminders, in-app notifications, and invoice lifecycle tracking (unpaid/paid/overdue) with indexed MongoDB queries for performant billing aggregation.
- Deployed on Vercel + Render with Winston structured logging, health check endpoint, and Jest unit tests for core handlers.

Banana Leaf Disease Detection: Enhanced ResNet50 | github.com/eej-sinining/Thesis-Banana-CNN · thesis-banana-cnn.onrender.com

Python 3.9 · TensorFlow 2.19 · TFLite · Flask · Docker · GitHub Actions · Unicorn

- Built a real-time banana leaf disease detection web app classifying Black Sigatoka, Cordana, and Healthy leaves using dual ResNet50 models with thread-safe TFLite inference (threading.Lock) served in parallel.
- Enhanced ResNet50 with Global Average Pooling (97% parameter reduction), AdamW + cosine annealing, and FP16 mixed precision — achieving ≥88.90% accuracy, 33% less training time (90→60 min), and 33% faster inference.
- Implemented multi-heuristic image validation to reject non-leaf inputs, occlusion-sensitivity saliency maps with disease bounding boxes, and 9-variant test-time augmentation for consistency scoring.
- Containerized with Docker, deployed on Render via GitHub Actions CI/CD, served behind Unicorn for production-grade reliability.

EDUCATION

Bachelor of Science in Computer Science 2022 – 2026

University of Mindanao, Davao City

TECHNICAL SKILLS

Languages: Java, Python, PHP, JavaScript, TypeScript, HTML5, CSS3

Frameworks: Spring Boot, Spring Security, React, Next.js, Laravel, Node.js

Data & Messaging: PostgreSQL, MySQL, MongoDB, Redis, Apache Kafka

DevOps & Security: Docker, GitHub Actions, Prometheus, Grafana, JWT, AES-256-GCM, OWASP

Tools: Git, Postman, Maven, Flyway, Figma, Jira

CERTIFICATIONS

- Information Technology Specialist in Databases — Certiport (2025)
- Information Technology Specialist in HTML and CSS — Certiport (2025)